JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON THURSDAY 22nd JANUARY 2026) **TIME : 3:00 PM TO 06:00 PM**

MATHEMATICS — TEST PAPER WITH ANSWER

SECTION-A

1. Let n be the number obtained on rolling a fair die. If the probability that the system

$x - ny + z = 6$

$x + (n-2)y + (n+1)z = 8$

$(n-1)y + z = 1$

Has a unique solution is $\frac{k}{6}$, then the sum of k and all possible values of n is :

(1) 21 (2) 24

(3) 20 (4) 22

2. If the mean deviation about the median of the numbers k, 2k, 3k, …., 1000k is 500, then k^2 is equal to :

(1) 16 (2) 4

(3) 1 (4) 9

3. The number of elements in the relation $R = \{(x,y): 4x^2 + y^2 < 52, x, y \in Z\}$ is

(1) 77 (2) 89

(3) 67 (4) 86

4. Let $S = \left\{z \in \mathbb{C} : 4z^2 + \bar{z} = 0\right\}$. Then $\sum_{z \in S} |z|^2$ is equal to :

(1) $\frac{3}{16}$ (2) $\frac{7}{64}$

(3) $\frac{1}{16}$ (4) $\frac{5}{64}$

5. If $\lim\limits_{x \to 0} \frac{e^{(a-1)x} + 2\cos bx + (c-2)e^{-x}}{x\cos x - \log_e(1+x)} = 2$, then $a^2 + b^2 + c^2$ is equal to :

(1) 5 (2) 3

(3) 7 (4) 9

6. If y = y(x) satisfies the differential equation

$16\left(\sqrt{x+9\sqrt{x}}\right)\left(4+\sqrt{9+\sqrt{x}}\right)\cos y\, dy = (1 + 2\sin y)dx$, $x > 0$ and $y(256) = \frac{\pi}{2}$, $y(49) = \alpha$, then $2\sin\alpha$ is equal to :

(1) $2\sqrt{2}-1$ (2) $2\left(\sqrt{2}-1\right)$

(3) $3\left(\sqrt{2}-1\right)$ (4) $\sqrt{2}-1$

7. Among the statements

(S1) : If A(5, –1) and B(–2, 3) are two vertices of a triangle, whose orthocentre is (0, 0), then its third vertex is (–4, –7) and

(S2) : If positive numbers 2a, b, c are three consecutive terms of an A.P., then the lines ax + by + c = 0 are concurrent at (2, –2),

(1) Only (S1) is correct

(2) Only (S2) is correct

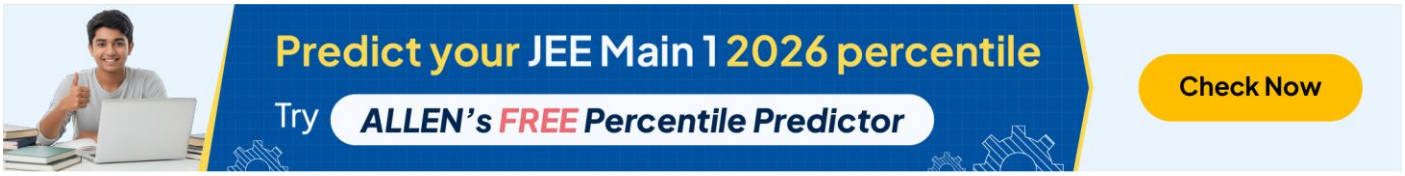
(3) Both are incorrect

(4) Both are correct

8. Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \lambda\hat{j} + 2\hat{k}$, $\lambda \in Z$ be two vectors, Let $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d}$ be a vector of magnitude 2 in yz-plane. If $|\vec{c}| = \sqrt{53}$, then the maximum possible value of $\left(\vec{c} \cdot \vec{d}\right)^2$ is equal to :

(1) 26 (2) 104

(3) 208 (4)52



9. If $X=\begin{bmatrix}x\\y\\z\end{bmatrix}$ is a solution of the system of equations $AX = B$, where adj $A=\begin{bmatrix}4&2&2\\-5&0&5\\1&-2&3\end{bmatrix}$ and $B=\begin{bmatrix}4\\0\\2\end{bmatrix}$, then $|x+y+z|$ is equal to :

(1) 3 (2) $\frac{3}{2}$

(3) 1 (4) 2

10. Let L be the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z+3}{6}$ and let S be the set of all points (a, b, c) on L, whose distance from the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z-9}{0}$ along the line L is 7. Then $\sum_{(a,b,c)\in S}(a+b+c)$ is equal to :

(1) 34 (2) 28

(3) 40 (4) 6

11. Let P $(10, 2\sqrt{15})$ be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, whose foci are S and S′. If the length of its latus rectum is 8, then the square of the area of $\Delta PSS'$ is equal to :

(1) 4200 (2) 900

(3) 1462 (4) 2700

12. The area of the region

$A=\{(x, y) : 4x^2+y^2\le 8 \text{ and } y^2\le 4x\}$ is :

(1) $\frac{\pi}{2}+2$ (2) $\pi+\frac{2}{3}$

(3) $\pi+4$ (4) $\frac{\pi}{2}+\frac{1}{3}$

13. Let α, β be the roots of the quadratic equation $12x^2-20x+3\lambda=0$, $\lambda\in\mathbb{Z}$. If $\frac{1}{2}\le|\beta-\alpha|\le\frac{3}{2}$, then the sum of all possible values of λ is :

(1) 6 (2) 1

(3) 3 (4) 4

14. Let the domain of the function

$$f(x)=\log_3\log_5\left(7-\log_2(x^2-10x+85)\right)+\sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$$

be $(\alpha, \beta]$. Then $\alpha+\beta$ is equal to :

(1) 10 (2) 12

(3) 9 (4) 8

15. Let [•] denote the greatest integer function, and let $f(x)=\min\left\{\sqrt{2}x, x^2\right\}$. Let $S=\{x\in(-2, 2) :$ the function $g(x)=|x|[x^2]$ is discontinuous at $x\}$.

Then $\sum_{x\in S}f(x)$ equals :

(1) $2-\sqrt{2}$ (2) $2\sqrt{6}-3\sqrt{2}$

(3) $1-\sqrt{2}$ (4) $\sqrt{6}-2\sqrt{2}$

16. Let S and S' be the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ and P (α, β) be a point on the ellipse in the first quadrant. If $(SP)^2+(S'P)^2-SP\bullet S'P=37$, then $\alpha^2+\beta^2$ is equal to :

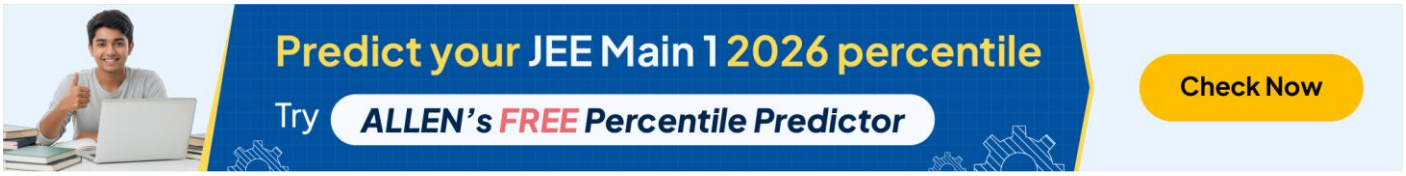
(1) 15 (2) 11

(3) 17 (4) 13

17. Let the locus of the mid-point of the chord through the origin O of the parabola $y^2=4x$ be the curve S. Let P be any point on S. Then the locus of the point, which internally divides OP in the ratio 3 : 1, is :

(1) $3y^2=2x$ (2) $2y^2=3x$

(3) $3x^2=2y$ (4) $2x^2=3y$



18. Let $f(x) = [x]^2 - [x+3] - 3$, $x \in \mathbb{R}$ where $[\bullet]$ is the greatest integer function. Then

(1) $f(x) > 0$ only for $x \in [4, \infty)$

(2) $f(x) < 0$ only for $x \in [-1, 3)$

(3) $\int_0^2 f(x)dx = -6$

(4) $f(x) = 0$ for finitely many values of x.

19. Let f and g be functions satisfying $f(x+y) = f(x)f(y)$, $f(1) = 7$ and $g(x+y) = g(xy)$, $g(1) = 1$, for all $x, y \in \mathbb{N}$. $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right) = 19607$, then n is equal to :

(1) 7 (2) 5

(3) 6 (4) 4

20. Let C_r denote the coefficient of x^r in the binomial expansion of $(1+x)^n$, $n \in \mathbb{N}$, $0 \le r \le n$.

If $P_n = C_0 - C_1 + \frac{2^2}{3}C_2 - \frac{2^3}{4}C_3 + ... + \frac{(-2)^n}{n+1}C_n$, then the value of $\sum_{n=1}^{25}\frac{1}{P_{2n}}$ equals.

(1) 580 (2) 525

(3) 650 (4) 675

SECTION-B

21. Let a vector $\vec{a} = \sqrt{2}\hat{i} - \hat{j} + \lambda\hat{k}$, $\lambda > 0$, make an obtuse angle with the vector $\vec{b} = -\lambda^2\hat{i} + 4\sqrt{2}\hat{j} + 4\sqrt{2}\hat{k}$ and an angle θ, $\frac{\pi}{6} < \theta < \frac{\pi}{2}$, with the positive z-axis. If the set of all possible values of λ is $(\alpha, \beta) - \{\gamma\}$, then $\alpha + \beta + \gamma$ is equal to _____.

22. Let $[\bullet]$ be the greatest integer function. If $\alpha = \int_0^{64}\left(x^{1/3} - \left[x^{1/3}\right]\right)dx$, then

$\frac{1}{\pi}\int_0^{\alpha\pi}\left(\frac{\sin^2\theta}{\sin^6\theta + \cos^6\theta}\right)d\theta$ is equal to ____.

23. Let $\cos(\alpha+\beta) = -\frac{1}{10}$ and $\sin(\alpha-\beta) = \frac{3}{8}$, where $0 < \alpha < \frac{\pi}{3}$ and $0 < \beta < \frac{\pi}{4}$.

If $\tan 2\alpha = \frac{3(1-r\sqrt{5})}{\sqrt{11}(s+\sqrt{5})}$, r, s $\in \mathbb{N}$, then r + s is equal to_____ .

24. Suppose a, b, c are in A.P. and a^2, $2b^2$, c^2 are in G.P. If $a < b < c$ and $a + b + c = 1$, then $9(a^2 + b^2 + c^2)$ is equal to_____ .

25. Let S be the set of the first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \ge 2$ and the product of all elements of B is even$\}$ is _____ .



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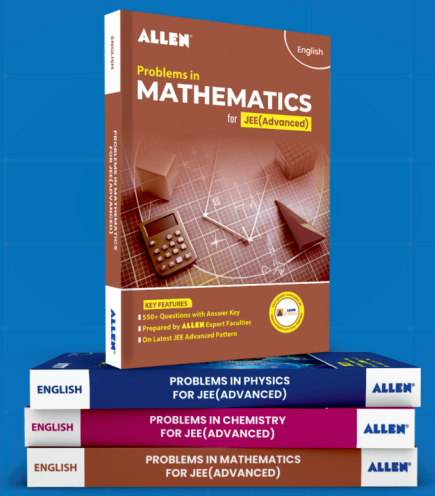


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